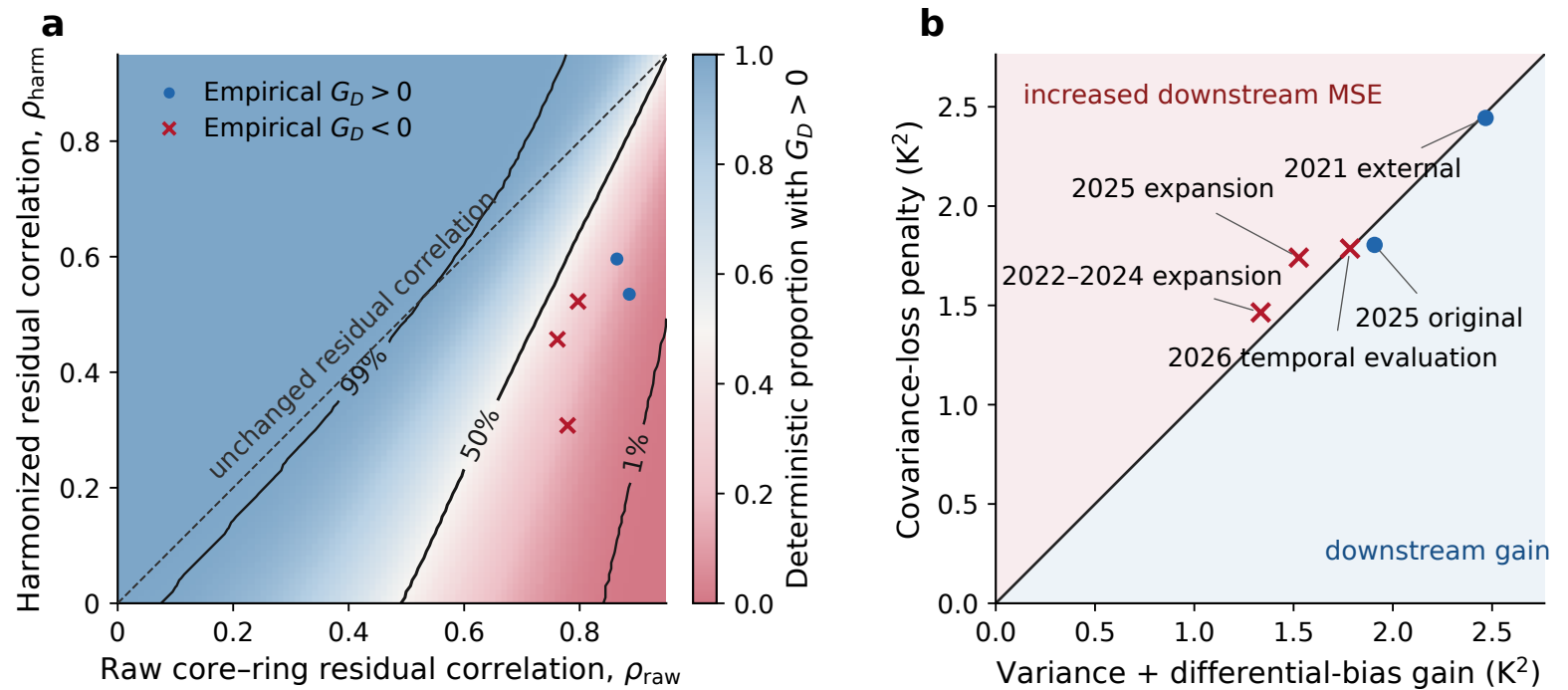
a
Empirical $G_D > 0$
Empirical $G_D < 0$
unchanged residual correlation
99%
50%
1%
Harmonized residual correlation, ρ_{harm}
Raw core–ring residual correlation, ρ_{raw}
Deterministic proportion with $G_D > 0$
b
increased downstream MSE
downstream gain
2021 external
2025 expansion
2022–2024 expansion
2025 original
2026 temporal evaluation
Covariance-loss penalty (K^2)
Variance + differential-bias gain (K^2)